

METHODS OF MODERN MATHEMATICAL PHYSICS

I: FUNCTIONAL ANALYSIS

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I: FUNCTIONAL ANALYSIS

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To

*R. S. Phillips and A. S. Wightman,
Mentors, Colleagues, Friends*

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Preface

This volume is the first of a three-volume series[†] devoted to an exposition of functional analysis methods in modern mathematical physics. It describes the fundamental principles of functional analysis and is essentially self-contained although there are occasional references to Volumes II and III. We have included a few applications when we thought that they would provide motivation for the reader. Volumes II and III describe various advanced topics in functional analysis and give numerous applications to modern physics. Throughout the three volumes we have included important applications of functional analysis techniques to classical physics and partial differential equations. The chapter titles following the table of contents give some idea of the topics in Volumes II and III.

It gives us pleasure to thank many individuals:

The students who took our course in 1970–1971 and especially J. E. Guckenheimer for constructive comments about the lectures and lecture notes.

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MIKE REED
BARRY SIMON

April 1972

[†] See Contents of Other Volumes, p. xvii, for revised contents.

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Introduction

Mathematics has its roots in numerology, geometry, and physics. Since the time of Newton, the search for mathematical models for physical phenomena has been a source of mathematical problems. In fact, whole branches of mathematics have grown out of attempts to analyze particular physical situations. An example is the development of harmonic analysis from Fourier's work on the heat equation.

Although mathematics and physics have grown apart in this century, physics has continued to stimulate mathematical research. Partially because of this, the influence of physics on mathematics is well understood. However, the contributions of mathematics to physics are not as well understood. It is a common fallacy to suppose that mathematics is important for physics only because it is a useful tool for making computations. Actually, mathematics plays a more subtle role which in the long run is more important. When a successful mathematical model is created for a physical phenomenon, that is, a model which can be used for accurate computations and predictions, the mathematical structure of the model itself provides a new way of thinking about the phenomenon. Put slightly differently, when a model is successful it is natural to think of the physical quantities in terms of the mathematical objects which represent them and to interpret similar or secondary phenomena in terms of the same model. Because of this, an investigation of the internal mathematical structure of the model can alter and enlarge our understanding of the physical phenomenon. Of course, the outstanding example of this is Newtonian mechanics which provided such a clear and coherent picture of celestial motions that it was used to interpret practically all physical phenomena. The model itself became central to an understanding of the physical world and it was difficult to give it up in the late nineteenth century, even in the face of contradictory evidence. A more modern example of this influence of mathematics on physics is the use of group theory to classify elementary particles.

The analysis of mathematical models for physical phenomena is part of the subject matter of mathematical physics. By analysis is meant both the rigorous derivation of explicit formulas and investigations of the internal mathematical structure of the models. In both cases the mathematical problems which arise lead to more general mathematical questions not associated with any particular model. Although these general questions are sometimes problems in pure mathematics, they are usually classified as mathematical physics since they arise from problems in physics.

Mathematical physics has traditionally been concerned with the mathematics of classical physics: mechanics, fluid dynamics, acoustics, potential theory, and optics. The main mathematical tool for the study of these branches of physics is the theory of ordinary and partial differential equations and related areas like integral equations and the calculus of variations. This classical mathematical physics has long been part of curricula in mathematics and physics departments. However, since 1926 the frontiers of physics have been concentrated increasingly in quantum mechanics and the subjects opened up by the quantum theory: atomic physics, nuclear physics, solid state physics, elementary particle physics. The central mathematical discipline for the study of these branches of physics is functional analysis, though the theories of group representations and several complex variables are also important. Von Neumann began the analysis of the framework of quantum mechanics in the years following 1926, but there were few attempts to study the structure of specific quantum systems (exceptions would be some of the work of Friedrichs and Rellich). This situation changed in the early 1950's when Kato proved the self-adjointness of atomic Hamiltonians and Gårding and Wightman formulated the axioms for quantum field theory. These events demonstrated the usefulness of functional analysis and pointed out the many difficult mathematical questions arising in modern physics. Since then the range and breadth of both the functional analysis techniques used and the subjects discussed in modern mathematical physics have increased enormously. The problems range from the concrete, for example how to compute or estimate the point spectrum of a particular operator, to the general, for example the representation theory of C^* -algebras. The techniques used and the general approach to the subject have become more abstract. Although in some areas the physics is so well understood that the problems are exercises in pure mathematics, there are other areas where neither the physics nor the mathematical models are well understood. These developments have had several serious effects not the least of which is the difficulty of communication between mathematicians and physicists. Physicists are often dismayed at the breadth of background and increasing mathematical sophistication which are required to understand the models. Mathematicians are often frustrated by

their own inability to understand the physics and the inability of physicists to formulate the problems in a way that mathematicians can understand.

A few specific remarks are appropriate. The prerequisite for reading this volume is roughly the mathematical sophistication acquired in a typical undergraduate mathematics education in the United States. Chapter I is intended as a review of background material. We expect that the reader will have some acquaintance with parts of the material covered in Chapters II–IV and have occasionally omitted proofs in these chapters when they seem uninspiring and unimportant for the reader.

The material in this book is sufficient for a two-semester course. Although we taught most of the material in a special one-semester course at Princeton which met five days a week, we do not recommend a repetition of that, either for faculty or students. In order that the material may be easily adapted for lectures, we have written most of the chapters so that the earlier sections contain the basic topics while the later sections contain more specialized and advanced topics and applications. For example, one can give students the basic ideas about unbounded operators in nine or ten lectures from Sections 1–4 of Chapter VIII. On the other hand, by doing the details of the proofs and by adding material from the notes and problems, Chapter VIII could easily become a one-semester course by itself.

Each chapter of this book ends with a long set of problems. Some of the problems fill gaps in the text (these are marked with a dagger). Others develop alternate proofs to the theorems in the text or introduce new material. We have also included harder problems (indicated by a star) in order to challenge the reader. We strongly encourage students to do the problems. It is trite but true that mathematics is learned by doing it, not by watching other people do it.

We hope that these volumes will provide physicists with an access to modern abstract techniques and that mathematicians will benefit by learning the advanced techniques side by side with their applications.

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Contents

<i>Preface</i>	<i>vii</i>
<i>Introduction</i>	<i>ix</i>
<i>Contents of Volumes II and III</i>	<i>xvii</i>

I: PRELIMINARIES

1. <i>Sets and functions</i>	<i>1</i>
2. <i>Metric and normed linear spaces</i>	<i>3</i>
<i>Appendix Lim sup and lim inf</i>	<i>11</i>
3. <i>The Lebesgue integral</i>	<i>12</i>
4. <i>Abstract measure theory</i>	<i>19</i>
5. <i>Two convergence arguments</i>	<i>26</i>
6. <i>Equicontinuity</i>	<i>28</i>
<i>Notes</i>	<i>31</i>
<i>Problems</i>	<i>32</i>

II: HILBERT SPACES

1. <i>The geometry of Hilbert space</i>	<i>36</i>
2. <i>The Riesz lemma</i>	<i>41</i>
3. <i>Orthonormal bases</i>	<i>44</i>
4. <i>Tensor products of Hilbert spaces</i>	<i>49</i>
5. <i>Ergodic theory: an introduction</i>	<i>54</i>
<i>Notes</i>	<i>60</i>
<i>Problems</i>	<i>63</i>

III: BANACH SPACES

1. <i>Definition and examples</i>	67
2. <i>Duals and double duals</i>	72
3. <i>The Hahn–Banach theorem</i>	75
4. <i>Operations on Banach spaces</i>	78
5. <i>The Baire category theorem and its consequences</i>	79
<i>Notes</i>	84
<i>Problems</i>	86

IV: TOPOLOGICAL SPACES

1. <i>General notions</i>	90
2. <i>Nets and convergence</i>	95
3. <i>Compactness</i>	97
<i>Appendix The Stone–Weierstrass theorem</i>	103
4. <i>Measure theory on compact spaces</i>	104
5. <i>Weak topologies on Banach spaces</i>	111
<i>Appendix Weak and strong measurability</i>	115
<i>Notes</i>	117
<i>Problems</i>	119

V: LOCALLY CONVEX SPACES

1. <i>General properties</i>	124
2. <i>Fréchet spaces</i>	131
3. <i>Functions of rapid decrease and the tempered distributions</i>	133
<i>Appendix The N-representation for $\mathcal{S}$ and $\mathcal{S}'$</i>	141
4. <i>Inductive limits: generalized functions and weak solutions of partial differential equations</i>	145
5. <i>Fixed point theorems</i>	150
6. <i>Applications of fixed point theorems</i>	153
7. <i>Topologies on locally convex spaces: duality theory and the strong dual topology</i>	162
<i>Appendix Polars and the Mackey–Arens theorem</i>	167
<i>Notes</i>	169
<i>Problems</i>	173

VI: BOUNDED OPERATORS

1. <i>Topologies on bounded operators</i>	182
2. <i>Adjoint</i> s	185
3. <i>The spectrum</i>	188
4. <i>Positive operators and the polar decomposition</i>	195
5. <i>Compact operators</i>	198
6. <i>The trace class and Hilbert–Schmidt ideals</i>	206
<i>Notes</i>	213
<i>Problems</i>	216

VII: THE SPECTRAL THEOREM

1. <i>The continuous functional calculus</i>	221
2. <i>The spectral measures</i>	224
3. <i>Spectral projections</i>	234
4. <i>Ergodic theory revisited: Koopmanism</i>	237
<i>Notes</i>	243
<i>Problems</i>	245

VIII: UNBOUNDED OPERATORS

1. <i>Domains, graphs, adjoints, and spectrum</i>	249
2. <i>Symmetric and self-adjoint operators: the basic criterion for self-adjointness</i>	255
3. <i>The spectral theorem</i>	259
4. <i>Stone’s theorem</i>	264
5. <i>Formal manipulation is a touchy business: Nelson’s example</i>	270
6. <i>Quadratic forms</i>	276
7. <i>Convergence of unbounded operators</i>	283
8. <i>The Trotter product formula</i>	295
9. <i>The polar decomposition for closed operators</i>	297
10. <i>Tensor products</i>	298
11. <i>Three mathematical problems in quantum mechanics</i>	302
<i>Notes</i>	305
<i>Problems</i>	312

<i>List of Symbols</i>	319
<i>Index</i>	321

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Contents of Other Volumes

Volume II: Fourier Analysis, Self-Adjointness

IX The Fourier Transform

X Self-Adjointness and the Existence of Dynamics

Volume III: Analysis of Operators

XI Perturbations of Point Spectra

XII Scattering Theory

XIII Spectral Analysis

Later Volumes

XIV Group Representations

XV Commutative Banach Algebras

XVI Convex Sets

XVII The GNS Construction

XVIII Von Neumann Algebras

XIX Applications to Quantum Field Theory

XX Applications to Statistical Mechanics